

RETAIL SALES ANALYSIS

An End-to-End Data Analytics Project

Python • Microsoft SQL Server • Power BI

9,994 orders analyzed | 2014-2017 | 4 regions | 3 categories

Executive Summary

This project analyzes retail sales transaction data from a superstore covering 9,994 orders (2014-2017) across multiple product categories, regions, and customer segments. The goal is to uncover where the business is profitable, where it is losing money, and why — to guide decisions on pricing, discounting, and regional strategy. The workflow covers data cleaning in Python, business analysis in SQL Server, and an interactive Power BI dashboard.

₹2.30M Total Sales	₹286K Total Profit	9,994 Total Orders
12.47% Profit Margin	18.72% Orders at a Loss	Technology Top Category

Key takeaway: Technology drives the most profit with the least discounting, while Furniture — especially Tables and Bookcases — is dragging overall profitability down despite steady sales. West region leads in profitability; Central lags.

1. Dataset Summary

- Rows: 9,994

Key Features:

- Order details (Order Date, Ship Date, Ship Mode, Order ID)
- Customer details (Customer ID, Customer Name, Segment)
- Location (Region, State, City)
- Product details (Category, Sub-Category, Product Name)
- Transaction metrics (Sales, Profit, Discount, Quantity)
- Missing Data: none found (checked with `df.isnull().sum()`)
- Duplicate rows: none found (checked with `df.duplicated().sum()`)

2. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

Data Loading: Imported the dataset using pandas.

Initial Exploration: Used `df.info()` and `df.describe()` to check structure and summary statistics.

Missing Data & Duplicate Check: Verified there were no missing values or duplicate rows in the dataset.

Column Standardization: Renamed columns to snake_case (lowercase, underscores) for readability and consistency.

Data Type Fixes: Converted `order_date` and `ship_date` to proper datetime format.

Feature Engineering:

- Created `order_year` and `order_month` columns from `order_date`
- Created `shipping_days` column (difference between `ship_date` and `order_date`)

Database Integration: Connected the cleaned DataFrame to Microsoft SQL Server for structured business-question analysis.

3. Data Analysis using SQL (Business Questions)

We performed structured analysis in Microsoft SQL Server to answer 12 key business questions:

1. Sales & Profit by Category — Technology leads in both sales and profit; Furniture has the highest sales-to-profit gap.

Category	Total Sales	Total Profit
Office Supplies	719,047.03	122,490.8
Furniture	741,999.8	18,451.27
Technology	836,154.03	145,454.95

2. Loss by Sub-Category — Tables is the single biggest loss-maker by a wide margin, followed by Bookcases and Supplies.

Sub-Category	Total Profit
Tables	-17,725.48
Bookcases	-3,472.56
Supplies	-1,189.1
Fasteners	949.52
Machines	3,384.76
Labels	5,546.25

3. Sales & Profit by Region — West leads in both sales and profit; Central has the lowest profit despite mid-range sales.

Region	Total Sales	Total Profit
West	725,457.82	108,418.45
East	678,781.24	91,522.78
South	391,721.91	46,749.43
Central	501,239.89	39,706.36

4. Monthly Sales Trend (2014-2017) — Sales grew year over year, with November/December consistently the strongest months (seasonal peak).

Year	Period	Sales (\$)
2014	Total (Jan-Dec)	410,868
2015	Total (Jan-Dec)	469,318
2016	Total (Jan-Dec)	608,073
2017	Total (Jan-Dec)	733,215
Peak month	November 2017	118,447.83

5. Top 10 Best-Selling Products — led by the Canon imageCLASS 2200 Advanced Copier.

Product ID	Product Name	Qty	Total Sales
TEC-CO-10004722	Canon imageCLASS 2200 Advanced Copier	20	61,599.82
OFF-BI-10003527	Fellowes PB500 Electric Punch Plastic Comb Binding...	31	27,453.38
TEC-MA-10002412	Cisco TelePresence System EX90 Videoconferencing...	6	22,638.48

Product ID	Product Name	Qty	Total Sales
FUR-CH-10002024	HON 5400 Series Task Chairs for Big and Tall	37	21,870.58
OFF-BI-10001359	GBC DocuBind TL300 Electric Binding System	37	19,823.48
OFF-BI-10005545	GBC Ibimaster 500 Manual ProClick Binding System	48	19,024.50
TEC-CO-10001449	Hewlett Packard LaserJet 3310 Copier	12	18,839.69
TEC-MA-10001127	HP Designjet T520 Inkjet Large Format Printer - 24"	12	18,374.90
OFF-BI-10004995	GBC DocuBind P400 Electric Binding System	27	17,965.07
OFF-SU-10000151	High Speed Automatic Electric Letter Opener	11	17,030.31

6. Top 10 Loss-Making Products — 3D printers and large conference/meeting tables account for the biggest losses.

Product ID	Product Name	Total Profit
TEC-MA-10000418	Cubify CubeX 3D Printer Double Head Print	-8,879.97
TEC-MA-10000822	Lexmark MX611dhe Monochrome Laser Printer	-4,589.97
TEC-MA-10004125	Cubify CubeX 3D Printer Triple Head Print	-3,839.99
FUR-TA-10000198	Chromcraft Bull-Nose Wood Oval Conference Tables	-2,876.12
FUR-TA-10001889	Bush Advantage Collection Racetrack Conference Table	-1,934.40
OFF-BI-10001359	GBC DocuBind P400 Electric Binding System	-1,878.17
TEC-MA-10002412	Cisco TelePresence System EX90 Videoconferencing	-1,811.08
OFF-SU-10002881	Martin Yale Chadless Opener Electric Letter Opener	-1,299.18
FUR-TA-10001950	Balt Solid Wood Round Tables	-1,201.06
FUR-TA-10004289	BoxOffice By Design Rectangular and Half-Moon Table	-1,148.44

7. Discount vs. Profit by Category — Furniture has the highest average discount (17.4%) and the lowest average profit per order; Technology has the lowest discount and highest profit.

Category	Avg Discount	Avg Profit	Total Profit
Furniture	17.39%	8.70	18,451.27
Office Supplies	15.73%	20.33	122,490.80
Technology	13.23%	78.75	145,454.95

8. Top 10 Customers by Sales — Sean Miller is the top spender by sales but is actually loss-making overall.

Customer ID	Customer Name	Total Sales	Total Profit
SM-20320	Sean Miller	25,043.05	-1,980.74
TC-20980	Tamara Chand	19,052.22	8,981.32
RB-19360	Raymond Buch	15,117.34	6,976.10
TA-21385	Tom Ashbrook	14,595.62	4,703.79
AB-10105	Adrian Barton	14,473.57	5,444.81
KL-16645	Ken Lonsdale	14,175.23	806.86
SC-20095	Sanjit Chand	14,142.33	5,757.41
HL-15040	Hunter Lopez	12,873.30	5,622.43
SE-20110	Sanjit Engle	12,209.44	2,650.68
CC-12370	Christopher Conant	12,129.07	2,177.05

9. Average Shipping Time by Ship Mode — as expected, Same Day is fastest and Standard Class slowest.

Ship Mode	Avg Shipping Days
Same Day	0 days
First Class	2 days
Second Class	3 days
Standard Class	5 days

10. Loss Order Percentage — roughly 1 in 5 orders results in a loss.

Loss Orders	Total Orders	Loss %
1,871	9,994	18.72%

11. Month-over-Month Sales Growth — volatile month to month, with a clear positive trend into Nov/Dec each year (sample below; full 48-month table available in the workbook).

Year	Month	Sales	Prev Month Sales	Growth %
2014	Jan	14,236.90	—	—
2014	Feb	4,519.89	11,951.41	-68.25%
2014	Nov	78,628.72	75,972.56	+3.50%
2015	Nov	75,972.56	44,261.11	+141.91%
2016	Dec	96,999.04	74,919.52	+29.47%
2017	Nov	118,447.83	77,776.92	+52.29%

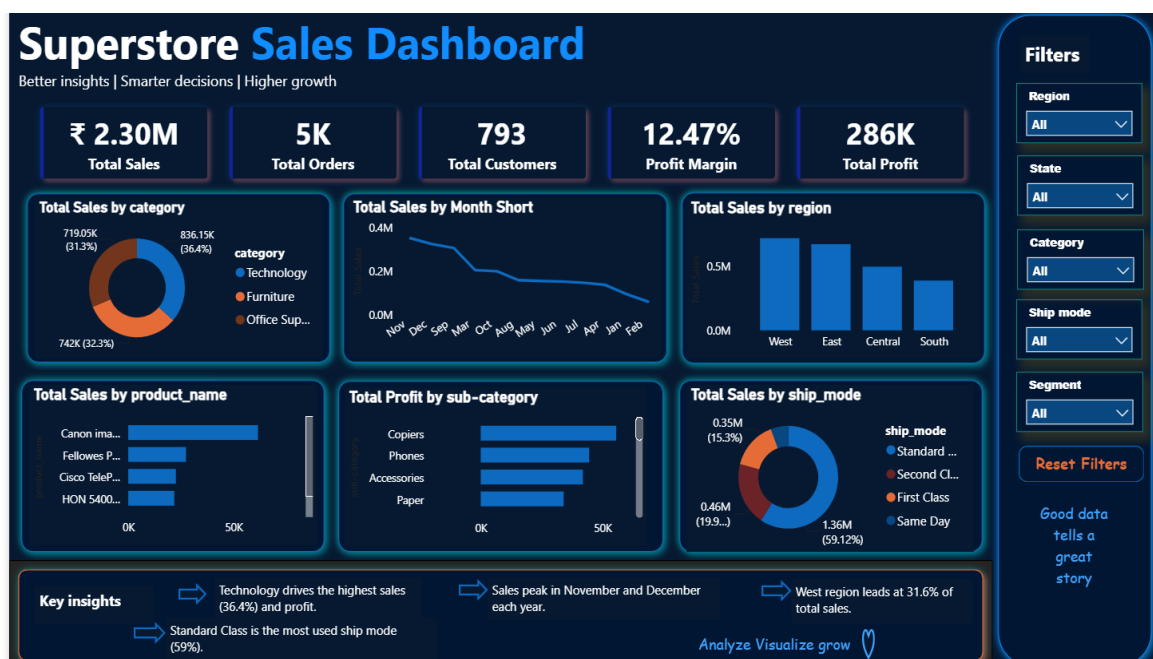
12. Ship Mode Impact on Sales/Profit — Standard Class handles ~60% of all orders and sales; First Class has the highest average profit per order.

Ship Mode	Total Orders	Total Sales	Total Profit	Avg Profit
Standard Class	5,968	1,358,215.73	164,088.79	27.49
Second Class	1,945	459,193.57	57,446.64	29.54
First Class	1,538	351,428.42	48,969.84	31.84
Same Day	543	128,363.13	15,891.76	29.27

4. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present these insights visually. The dashboard includes:

- 5 KPI cards: Total Sales (₹2.30M), Total Orders (5K), Total Customers (793), Profit Margin (12.47%), Total Profit (286K)
- Sales by Category (donut chart)
- Sales trend by month (line chart)
- Sales by Region (bar chart)
- Top products by sales (bar chart)
- Profit by Sub-Category (bar chart, highlighting loss-making sub-categories)
- Sales by Ship Mode (donut chart)
- Filters: Region, State, Category, Ship Mode, Segment
- Key Insights strip summarizing the top findings



5. Business Recommendations

Review Discount Policy on Furniture — Furniture has the highest average discount (17.4%) and lowest average profit per order; Tables and Bookcases are net loss-makers overall. Capping discounts on these sub-categories could recover meaningful profit.

Investigate 3D Printer & Large Table Pricing — The single biggest loss-making products are 3D printers (Cubify CubeX) and large conference/meeting tables — these may be underpriced relative to cost or over-discounted.

Double Down on Technology — Technology has the lowest average discount (13.2%) and by far the highest profit per order (₹78.75 avg) — a strong candidate for continued inventory and marketing investment.

Improve Central Region Profitability — Central region has the lowest profit despite mid-range sales, suggesting higher costs or heavier discounting there — worth a deeper regional cost review.

Plan for Seasonality — Sales consistently peak in November and December each year; ensure inventory and staffing are scaled up ahead of this period.

Review High-Spend, Low-Profit Customers — Customers like Sean Miller generate high sales but negative profit overall — worth reviewing the discount terms extended to top accounts.

6. Conclusion

This project demonstrates a complete analytics workflow: cleaning and preparing raw data in Python, answering targeted business questions in SQL, and communicating results through an interactive Power BI dashboard. The analysis moved beyond simple sales reporting to identify specific, actionable drivers of profitability — which sub-categories, regions, and discount practices are helping or hurting the business.